- $\Phi : \mathcal{L}(\mathcal{H}_1) \longrightarrow \mathcal{L}(\mathcal{H}_2)$ QUANTUM CHANNEL $\overset{\text{DEF}}{\iff}$
  - LINEAR
    $\Phi(\alpha \rho_1 + \beta \rho_2) = \alpha \Phi(\rho_1) + \beta \Phi(\rho_2)$
  - COMPLETELY POSITIVE
    - $A \in \mathcal{H}_1 \otimes \mathcal{H}_E$
    - $A \geq 0$
    $\Rightarrow (\Phi \otimes \mathbb{1})(A) \geq 0$
  - TRACE PRESERVING.
    $\mathrm{tr}(\Phi(X)) = \mathrm{tr}(X)$

- DEF: CHOI-STATE of $\Phi(\cdot) : \mathcal{L}(\mathcal{H}_A) \to \mathcal{L}(\mathcal{H}_A)$

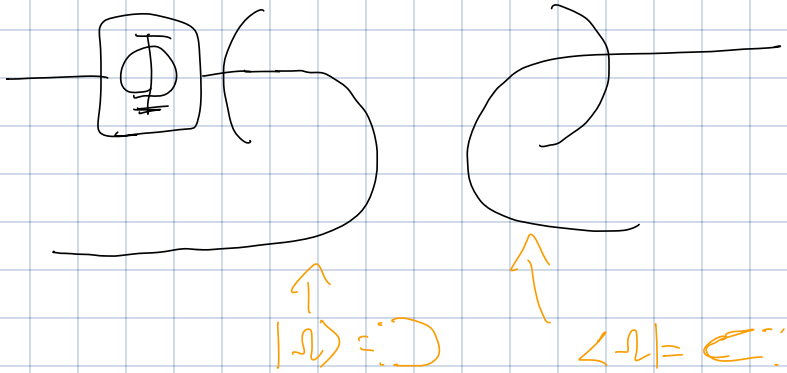
$\mathcal{H}_B$ space of same dimension d_B (orthonormal basis states)

$$\rho_\Phi := (\Phi \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|) \quad \text{with} \quad |\Omega\rangle := \sum_{i=1}^{d} \frac{1}{d} |i\rangle \otimes |i\rangle$$

We have $\rho_\Phi = \frac{1}{d}(\Phi \otimes \mathbb{1}) \left(\sum_{i,j}^{d} |ii\rangle\langle jj| \right) = \frac{1}{d} \sum_{i,j=1}^{d} \Phi(|i\rangle\langle j|) \otimes |i\rangle\langle j|$

- Using diagrams a Choi state is:

$\rho_\Phi = \frac{1}{d}$ ·



We have $\rho_\Phi \geq 0 \iff \Phi(\cdot)$ is completely positive.

PROOF:

"$\Leftarrow$" Easy: $\rho_\Phi := \Phi \otimes \mathbb{1}(|\Omega\rangle\langle\Omega|) \geq 0$
(Φ compl. positive)

"$\Rightarrow$": I have to show: $\rho_\Phi \geq 0 \Rightarrow \left(\sigma \geq 0 \Rightarrow (\Phi \otimes \mathbb{1})(\sigma) \geq 0 \right)$

$$\Phi \otimes \mathbb{1}(\sigma) = \sum_{i}^{d_A} \lambda_i (\Phi \otimes \mathbb{1})(|v_i\rangle\langle v_i|)$$

$\dim(\mathcal{H}_B) \geq \dim(\mathcal{H}_A)$

$\sigma = \sum_{i=1}^{d_A} \lambda_i |v_i\rangle\langle v_i|$, $\lambda_i \geq 0$

Let's focus on $(\Phi \otimes \mathbb{1})(|V_i\rangle\langle V_i|)$.

$d_B \geq d_A \Rightarrow \min(d_A, d_B) = d_A$

- $|V_i\rangle = \sum_{j=1}^{\min(d_A,d_B)} s_j |j\rangle \otimes |j\rangle = \sum_{j=1}^{\min(d_A,d_B)} X \otimes \mathbb{1} |j\rangle \otimes |j\rangle$ ($s_j \geq 0$)
  - $|V_i\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$
  - SCHMIDT DECOMP.

  ($\exists X$ such that $X|j\rangle = s_j |j\rangle$)

$$= (X \otimes \mathbb{1}) W_A \otimes W_B |\Omega\rangle$$

- Where W_A and W_B are unitary such that: $\sum_j (W_A^\dagger |j\rangle) \otimes (W_B^\dagger |j\rangle) = |\Omega\rangle$

$$= X \otimes \mathbb{1} (\mathbb{1} \otimes W_B W_A^\dagger) |\Omega\rangle =$$

(TRANSPOSE-TRICK)

$$= \mathbb{1} \otimes W_B W_A^\dagger X^\dagger |\Omega\rangle$$

- $|V_i\rangle\langle V_i| = (\mathbb{1} \otimes W_B W_A^T X^T) |\Omega\rangle\langle\Omega| (\mathbb{1} \otimes (W_B W_A^\dagger X^\dagger)^+)$
- $(\Phi \otimes \mathbb{1})[|V_i\rangle\langle V_i|] = \underbrace{(\mathbb{1} \otimes W_B W_A^T X^T)}_{M} (\Phi \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|) (\mathbb{1} \otimes (W_B W_A^\dagger X^\dagger)^+)$

$$= M P_\Phi M^+$$

Now $\forall |\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ $\quad \langle\psi| M P_\Phi M^+ |\psi\rangle = \langle M^\dagger \psi| P_\Phi |M^\dagger \psi\rangle$

$$= \langle v | P_\Phi | v \rangle \geq 0$$

$P_\Phi \geq 0$ (ASSUMPTION.)

- Now we have proven that $P_\Phi \geq 0 \Rightarrow \Phi \otimes \mathbb{1} \geq 0$ ($\mathbb{1}$ on $\mathcal{H}_B$ with $\dim(\mathcal{H}_B) = \dim(\mathcal{H}_A)$)

- If $\dim(H_B) < \dim(H_A)$:

$$\phi \otimes \mathbb{1}_B(V) = \mathrm{tr}_{B'}\left(\phi \otimes \underbrace{\mathbb{1}_B \otimes \mathbb{1}_{B'}}_{\text{same dimension of } A \text{ in total}}(V \otimes |0\rangle_{B'}\langle 0|_{B'})\right) \geq 0 \quad \Rightarrow \square$$

- ρ_Φ uniquely define Φ using:

$$\Phi(V) = d\,\mathrm{tr}_B\left(\mathbb{1} \otimes V^t \; \rho_\Phi\right)$$

PROOF:

$$d\,\mathrm{tr}_B\left(\mathbb{1} \otimes V^t \; \rho_\Phi\right) = \frac{d}{d}\mathrm{tr}_B\left(\mathbb{1} \otimes V^t \sum_{i,j} \phi(|i\rangle\langle j|) \otimes |i\rangle\langle j|\right) =$$

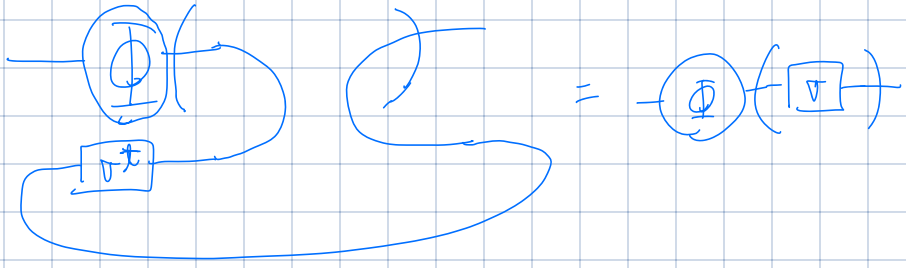
$$= \sum_{i,j} \mathrm{tr}_B\left(\phi(|i\rangle\langle j|) \otimes V^t |i\rangle\langle j|\right) =$$

$$= \sum_{i,j} \phi(|i\rangle\langle j|)\langle j|V^t|i\rangle =$$

$$= \sum_{i,j} \phi(|i\rangle\langle j|)\langle i|V|j\rangle = \phi(V)$$

($\sum |i\rangle\langle i| = \mathbb{1}$)

Or Using diagrams:



- $\mathrm{tr}_A(\rho_\phi) = \frac{\mathbb{1}_B}{d}$ ← Only the state ρ such that $\mathrm{tr}_A(\rho) = \frac{1}{d}$ can be Choi state.

PROOF:

$$\mathrm{tr}_A\left(\Phi \otimes \mathbb{1}(|\Omega\rangle\langle\Omega|)\right) = \frac{1}{d}\mathrm{tr}_A\left(\sum_{i,j}\phi(|i\rangle\langle j|) \otimes |i\rangle\langle j|\right) = \frac{1}{d}\sum_{i,j}\mathrm{tr}\left(\phi(|i\rangle\langle j|)\right)|i\rangle\langle j| =$$

$$= \frac{1}{d}\sum_{i,j}\mathrm{tr}(|i\rangle\langle j|)\,|i\rangle\langle j| = \frac{1}{d}\sum_{i}|i\rangle\langle i| = \frac{\mathbb{1}}{d}$$

(ϕ trace preserving)

THEOREM KRAUS

- Φ quantum channel $\iff$ $\exists \{K_i\}_{i=1}^{d^2} \in \mathcal{L}(H)$ such that:

  (KRAUS OPERATORS)

$$\begin{cases} \cdot\ \Phi(\rho) = \sum_{i=1}^{d^2} K_i \rho K_i^+ \\ \cdot\ \sum_{i=1}^{d^2} K_i^+ K_i = \mathbb{1} \end{cases}$$

PROOF:

"$\Leftarrow$"

- Linear : easy
- Trace preserving : $Tr(\Phi(\rho)) = \sum_{i=1}^{d^2} Tr(K_i \rho K_i^+) = \sum_i Tr(K_i^+ K_i \rho) =$

$$= Tr\left(\underbrace{\sum_i K_i^+ K_i}_{\mathbb{1}} \rho\right) = Tr(\rho)$$

- Completely positive: $\sigma \geq 0 \Rightarrow \forall |\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B \quad \langle\psi| \Phi \otimes \mathbb{1}(\sigma) |\psi\rangle \geq 0$

SUBPROOF:

$$\langle\psi|(\Phi \otimes \mathbb{1})(\sigma)|\psi\rangle = \langle\psi| \sum_i K_i \otimes \mathbb{1}\ \sigma\ K_i^+ \otimes \mathbb{1} |\psi\rangle =$$

$$= \sum_i \underbrace{\langle\psi|(K_i \otimes \mathbb{1})}_{\langle v_i|}\ \sigma\ \underbrace{(K_i^+ \otimes \mathbb{1})|\psi\rangle}_{|v_i\rangle}$$

$$= \sum_i \langle v_i|\sigma|v_i\rangle \geq 0 \quad (\sigma \geq 0)$$

"$\Rightarrow$"

- $\Phi(\rho) = \sum_{i=1}^{d} \lambda_i \Phi(|v_i\rangle\langle v_i|)$ ($\rho = \rho^+ \Rightarrow$ eigendec.)
- $\Phi(|\psi\rangle\langle\psi|) = d\, {}_B\langle\psi^*| P_\Phi |\psi^*\rangle_B \equiv$ ★

SUB PROOF:

$${}_B\langle\psi^*| P_\Phi |\psi^*\rangle_B = {}_B\langle\psi^*| (\Phi \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|) |\psi^*\rangle_B =$$

$$= \frac{1}{d} \sum_{i,j} \Phi(|i\rangle_A\langle j|)\ \langle\psi^*|i\rangle\langle j|\psi^*\rangle$$

$$\left(\Phi \otimes \mathbb{1}_B(|\Omega\rangle\langle\Omega|) = \frac{1}{d} \sum_{i,j} \Phi(|i\rangle_A\langle j|) \otimes |i\rangle_B\langle j| \right)$$

$$= \frac{1}{d} \sum_{i,j} \Phi\left(\langle i|\psi\rangle |i\rangle \langle j| \langle \psi|j\rangle \right) = \frac{1}{d} \Phi\left(|\psi\rangle\langle\psi| \right)$$

$\Rightarrow$ □ SUBPROOF.

$\left(\langle \psi^*|i\rangle = (\langle i|\psi^*\rangle)^* = \langle i^*|\psi\rangle = \langle i|\psi\rangle \right.$

$\left. \begin{pmatrix} 0 \\ 1 \\ \vdots \end{pmatrix} = |i\rangle \text{ comp. basis.} \right)$

- $\Phi(\rho) = \sum_{i=1}^{d} \lambda_i \Phi\left(|v_i\rangle\langle v_i| \right) = \sum_{i=1}^{d} \lambda_i \langle v_i^*|_B P_\Phi |v_i^*\rangle_B d$

$$= \sum_{i=1}^{d} \lambda_i \sum_{j=1}^{d^2} p_j \langle v_i^*|\psi_j\rangle_{AB} \langle \psi_j| v_i^*\rangle_B d =$$

$\left(P_\Phi \geq 0 \Rightarrow P_\Phi^\dagger = P_\Phi \Rightarrow \text{diagonalize.} \right.$

$\left. \Rightarrow P_\Phi = \sum_{j=1}^{d^2} p_j |\psi_j\rangle_{AB} \langle \psi_j| \right)$

$$= \sum_{j=1}^{d^2} p_j d \sum_{i=1}^{d} \lambda_i \langle v_i^*|\psi_j\rangle_{AB} \langle \psi_j| v_i^*\rangle_B =$$

$$= \sum_{j=1}^{d^2} \sum_{i=1}^{d} \lambda_i K_j |v_i\rangle\langle v_i| K_j^\dagger =$$

$\left(K_j |\bullet\rangle_A := \langle \bullet^*|\psi_j\rangle_{AB} \sqrt{p_j} \sqrt{d} \right.$

$\left. \text{linear operator} \right)$

$$= \sum_{j=1}^{d^2} K_j \rho K_j^\dagger$$

$\sum_{i=1}^{d} \lambda_i |v_i\rangle\langle v_i| = \rho$

- $\operatorname{tr}(\Phi(\rho)) = 1 \Rightarrow \operatorname{tr}\left(\sum_j K_j^\dagger K_j \rho \right) = 1 \quad \forall \rho \Rightarrow \sum_{j=1}^{d^2} K_j^\dagger K_j = \mathbb{1}$

STINESPRING REPRESENTATION.

$\Phi(\cdot)$ quantum channel $\Longleftrightarrow$ $\exists\ U_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ (auxiliary system) unitary such that

$$\Phi(\rho) = \mathrm{tr}_B\left(U_{AB}\, \rho \otimes |0\rangle\langle 0|\, U_{AB}^\dagger \right)$$

PROOF:

"$\Longrightarrow$" Φ q. channel $\Leftrightarrow$ $\exists\ \{K_i\}_{i=1}^{d^2}$ such that $\Phi(\rho) = \sum_{i=1}^{d^2} K_i \rho K_i^\dagger$ and $\sum_i K_i^\dagger K_i = \mathbb{1}$

CLAIM: $U_{AB}|0\rangle \otimes |0\rangle = \sum_{i=1}^{d^2} K_i |\psi\rangle \otimes |i\rangle$

(U_{AB}'s unitary)

SUBPROOF:

$$U_{AB}\, \rho \otimes |0\rangle\langle 0|\, U_{AB}^\dagger = \sum_i \lambda_i\, U_{AB}(|v_i\rangle \otimes |0\rangle)(\langle v_i| \otimes \langle 0|) U_{AB}^\dagger =$$

($\rho = \sum_i \lambda_i |v_i\rangle\langle v_i|$)

$$= \sum_i \lambda_i \sum_{j=1}^{d^2} (K_j |v_j\rangle \otimes |j\rangle) \sum_{\ell=1}^{d^2} (\langle v_\ell | K_\ell^\dagger \otimes \langle \ell|)$$

$$\mathrm{tr}_B\left(U_{AB}\, \rho \otimes |0\rangle\langle 0|\, U_{AB}^\dagger \right) = \sum_i \lambda_i \sum_{j=1}^{d^2} \sum_{\ell=1}^{d^2} (K_j |v_j\rangle\langle v_\ell| K_\ell^\dagger)\, \underbrace{\mathrm{tr}_B(|j\rangle\langle \ell|)}_{=\delta_{j,\ell}}$$

$$= \sum_{j=1}^{d^2} K_j \rho K_j^\dagger$$

($\sum \lambda_i |v_i\rangle\langle v_i| = \rho$)

"$\Longleftarrow$" $\Phi(\rho) = \mathrm{tr}_B\left(U_{AB}\, \rho \otimes |0\rangle\langle 0|\, U_{AB}^\dagger \right)$ is:

- linear: Yes!
- trace preserving: Yes! as $\mathrm{tr}(\Phi(\rho)) = \ldots = 1$
- Complet. positive: $(\Phi \otimes \mathbb{1})(\nabla) = \mathrm{tr}_B\left(U_{AB} \nabla \otimes |0\rangle\langle 0|\, U_{AB}^\dagger \right) \geq 0.$

($\mathbb{1}$ on $\mathcal{H}_L$; ∇ on $\mathcal{H}_A \otimes \mathcal{H}_L$; U_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_L$)